AI-BASED MULTIMODAL SYSTEM FOR EARLY AUTISM SPECTRUM DISORDER SCREENING

PROJECT REPORT

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CERTIFICATE

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ABSTRACT

Autism Spectrum Disorder (ASD) has emerged as a significant global health concern due to its increasing prevalence and the critical importance of early diagnosis. It is a neurodevelopmental condition characterized by challenges in social interaction, communication, and repetitive behavioral patterns. Early identification and timely intervention are essential for improving developmental outcomes and quality of life for individuals with ASD. However, existing diagnostic approaches largely depend on clinical observation and expert evaluation, which are often time-consuming, subjective, and not easily accessible in all regions. These limitations can delay early screening and reduce the chances of effective intervention.

To overcome these challenges, there is a growing need for efficient, accessible, and technology-driven screening solutions. The proposed system addresses this need by introducing an AI-based multimodal framework for ASD screening. It integrates a structured behavioral questionnaire with additional data-driven techniques to enhance assessment reliability. The behavioral module collects user responses related to social interaction, communication patterns, and repetitive behaviors, which are analyzed using machine learning models to estimate the likelihood of ASD traits. Furthermore, the system incorporates visual-based analysis, including facial and gaze-related features, to strengthen the screening process.

The application is developed as a web-based platform with an interactive user interface that enables users to input data and obtain results in real time. The backend processes the inputs, applies trained models, and generates screening insights. By combining multiple modalities into a unified system, the framework aims to improve screening accuracy while remaining user-friendly and scalable. This approach serves as a supportive tool for preliminary assessment and encourages timely professional consultation, ultimately contributing to increased awareness and early intervention in autism care.

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CHAPTER 1: INTRODUCTION

1.1 OVERVIEW

Autism Spectrum Disorder (ASD) is a complex neurodevelopmental disorder that affects an individual's ability to communicate, interact socially, and exhibit appropriate behavioral responses. It is characterized by a wide spectrum of symptoms, including difficulties in verbal and non-verbal communication, repetitive behaviors, restricted interests, and impaired social interactions. The severity and manifestation of these symptoms vary significantly among individuals, making ASD a heterogeneous condition.

Over the past decade, there has been a noticeable increase in the number of ASD cases reported worldwide. Early identification of ASD plays a crucial role in improving developmental outcomes, as timely intervention can significantly enhance cognitive, social, and behavioral skills. However, many children are diagnosed at later stages due to limitations in existing diagnostic methods.

Traditional diagnostic approaches primarily rely on clinical observation, behavioral assessments, and structured questionnaires administered by trained professionals. These methods are often time-consuming, subjective, and dependent on expert availability. Additionally, such diagnostic processes may not be easily accessible in rural or under-resourced regions, leading to delays in detection and intervention.

With advancements in Artificial Intelligence (AI) and Machine Learning (ML), there is a growing interest in developing automated systems that can assist in early ASD screening. These systems leverage computational techniques to analyze behavioral, visual, and cognitive indicators, thereby providing objective and efficient assessments.

In this project, a multimodal autism screening system is proposed, which integrates multiple data sources such as facial analysis, eye gaze tracking, and behavioral questionnaires. By combining these modalities, the system aims to provide a comprehensive and reliable assessment of autism risk. The proposed system is designed as a web-based platform, making it accessible, scalable, and user-friendly for parents, educators, and healthcare professionals.

1.2 MOTIVATION

The motivation for developing this project arises from the increasing need for early, accurate, and accessible autism screening solutions. Several key factors contribute to this motivation.

Firstly, early diagnosis of ASD is essential for effective intervention. Studies have shown that children diagnosed at an early age benefit significantly from targeted therapies, leading to improved developmental outcomes. However, many cases are identified only after noticeable behavioral issues emerge, which may delay treatment.

Secondly, the availability of trained specialists such as child psychologists and neurologists is limited, especially in rural and economically disadvantaged regions. This creates a significant gap between the demand for diagnosis and the availability of expert services.

Thirdly, existing diagnostic methods are often expensive and time-intensive. Families may need to undergo multiple consultations and assessments, which increases both financial and emotional burden.

Another important motivation is the subjectivity involved in traditional diagnostic approaches. Human observation and interpretation can vary, leading to inconsistencies in diagnosis. This highlights the need for an objective and standardized screening system. Additionally, data privacy and security are crucial in such systems, as sensitive user information must be protected through secure storage and processing mechanisms. Ensuring proper user consent and ethical handling of data further enhances the trustworthiness and acceptance of the solution.

Furthermore, recent advancements in deep learning, computer vision, and data analysis techniques provide an opportunity to develop intelligent systems capable of identifying subtle behavioral patterns associated with ASD. These technologies can be utilized to automate the screening process and improve accuracy.

Therefore, this project aims to develop a cost-effective, scalable, and AI-based solution that can assist in early autism screening, thereby reducing diagnosis time, cost, and dependency on expert availability.

1.3 CHALLENGES IN ASD SCREENING

Despite significant progress in technology, several challenges exist in designing an effective autism screening system.

One of the major challenges is the variability of ASD symptoms. Since ASD manifests differently in each individual, it is difficult to create a single model that accurately captures all behavioral patterns. This variability requires the system to be flexible and capable of handling diverse inputs.

Another challenge is the limited availability of high-quality datasets. Collecting labeled data for ASD detection involves ethical considerations, privacy concerns, and the need for expert validation. This often results in smaller datasets, which can affect model performance.

Single-modality approaches, which rely on only one type of input such as facial images or questionnaires, are often insufficient for accurate detection. Such systems may miss important behavioral cues, leading to reduced reliability.

Integrating multiple modalities presents its own challenges. Different data types such as images, gaze patterns, and questionnaire responses require different processing techniques. Combining these heterogeneous data sources into a unified model is a complex task.

Computational complexity is another concern. Advanced machine learning models, particularly deep learning architectures, require significant computational resources for training and inference. Ensuring that the system remains efficient and responsive in a web-based environment is a critical challenge. Ensuring data privacy, ethical compliance, and secure handling of sensitive user information further adds to the complexity of developing a reliable and widely acceptable autism screening system.

Additionally, ensuring real-time performance for components such as gaze tracking requires optimized algorithms and efficient implementation.

Addressing these challenges is essential for developing a robust and reliable autism screening system.

1.4 OBJECTIVES

The primary objective of this project is to design and develop a multimodal, AI-based system for the early screening of Autism Spectrum Disorder (ASD) in children. The system aims to assist in identifying potential signs of autism at an early stage, enabling timely intervention and support.

A key goal of the project is to build a web-based platform that is both accessible and user-friendly. The interface is designed so that parents, caregivers, and professionals can easily interact with the system without requiring technical expertise.

The project focuses on integrating multiple modalities, including facial analysis, eye gaze tracking, and behavioral questionnaires. By combining these different sources of information, the system aims to provide a more comprehensive and reliable screening process compared to single-method approaches.

Another important objective is to implement machine learning models capable of analyzing facial features and identifying patterns commonly associated with ASD. These models help in detecting subtle cues that may not be easily observable through manual assessment.

In addition, the system incorporates a gaze tracking mechanism to study attention patterns and social interaction behavior. This helps in understanding how children respond to visual stimuli, which is an important indicator in autism screening.

The project also includes standardized behavioral questionnaires to collect structured input from caregivers. These questionnaires play a crucial role in capturing developmental and behavioral traits that complement the technical analysis.

To improve accuracy, the outputs from different modalities are combined using an effective fusion technique. This integrated approach ensures that the final assessment is more balanced and reduces reliance on any single data source.

The system is designed to generate an automated report that indicates the potential risk level of autism. This report provides clear and understandable results that can assist in decision-making for further clinical evaluation.

Finally, the project aims to reduce the time, cost, and subjectivity involved in traditional diagnostic methods. It is also developed with scalability in mind, allowing future enhancements and adaptability as new technologies and research findings emerge.

1.5 SCOPE OF THE PROJECT

The scope of this project is focused on developing an automated screening tool for early detection of autism risk. The system is intended to assist parents, teachers, and healthcare professionals in identifying potential signs of ASD at an early stage.

The proposed system includes the development of a web-based interface through which users can provide input data such as images, gaze interactions, and questionnaire responses. The system processes these inputs using machine learning models and generates a risk assessment.

The application is designed to be accessible through standard devices such as laptops, tablets, and smartphones, ensuring usability in both urban and rural settings.

It is important to note that the system is intended for screening purposes only and does not replace professional medical diagnosis. Instead, it serves as a decision-support tool that can guide users toward seeking further clinical evaluation if necessary.

The system is also designed with scalability in mind, allowing for the integration of additional features such as speech analysis, behavioral tracking, and multilingual support in future developments.

CHAPTER 2: LITERATURE REVIEW

Autism Spectrum Disorder (ASD) has become a major area of research due to its increasing prevalence and the importance of early diagnosis. Traditional diagnostic techniques rely on behavioral observation and clinical expertise, which are often time-consuming and subjective. To overcome these limitations, researchers have explored computational approaches using machine learning (ML), deep learning (DL), and multimodal analysis.

Recent studies have focused on analyzing behavioral, visual, and cognitive patterns using advanced algorithms. These include facial image analysis, eye gaze tracking, speech analysis, and questionnaire-based models. This chapter reviews the major research contributions in ASD detection and highlights their methodologies, findings, and limitations.

Despite these advancements, several challenges still remain in achieving reliable and widely applicable ASD screening systems. Many existing approaches rely on single-modality data, which may limit accuracy and fail to capture the complexity of autism-related behaviors. Additionally, issues such as limited dataset availability, lack of real-world validation, and difficulties in integrating multiple data sources continue to affect performance. These limitations highlight the need for more robust, scalable, and multimodal solutions that can provide consistent and practical support for early ASD screening..

2.1 MACHINE LEARNING-BASED ASD DETECTION

Machine learning techniques have been widely used for ASD prediction using structured datasets such as behavioral questionnaires and medical records. These models typically use classifiers such as Support Vector Machines (SVM), Logistic Regression, and Random Forest.

A study by Farooq et al. (2023) applied machine learning models combined with federated learning to detect ASD across multiple datasets. The approach enabled decentralized data processing and achieved reliable prediction accuracy while preserving data privacy. Similarly, Wu et al. (2021) developed a machine learning system that analyzes behavioral features extracted from infant videos. The model focused on behaviors such as gaze direction, emotional expression, and vocalization, demonstrating the effectiveness of machine learning in early ASD detection.

Overall, the advantages of machine learning approaches include simplicity, computational efficiency, and strong performance on structured datasets, making them easy to deploy in practical systems. However, the limitations include reliance on manually extracted features and the inability to effectively capture complex visual or temporal behavioral patterns, which may reduce performance in real-world scenarios.

2.2 FACIAL IMAGE BASED ASD DETECTION

Facial analysis has become an important area of research due to its non-invasive nature. It enables the examination of visual facial cues without requiring physical contact or intrusive procedures, making it suitable for large-scale and real-world screening applications. Deep learning models, particularly Convolutional Neural Networks (CNNs), have been widely applied for this purpose because of their strong capability to automatically learn hierarchical features from images. These models can capture complex spatial patterns such as facial geometry, symmetry, and texture variations that may not be easily identifiable through manual feature extraction. Additionally, the availability of pre-trained models and transfer learning techniques has further improved performance by allowing models to leverage knowledge from large-scale image datasets. This approach reduces training time while enhancing generalization, especially in domains with limited labeled medical data.A study by Ahmad et al. (2024) evaluated multiple pre-trained CNN models such as ResNet, VGG, and MobileNet for ASD detection using facial images. Facial analysis systems can detect subtle differences in facial expressions and emotional responses, which are important indicators of ASD. Advanced models are also capable of analyzing micro-expressions and emotional recognition patterns.

In summary, the advantages of facial image-based methods include their non-invasive nature, high accuracy with deep learning models, and ability to detect subtle behavioral cues. However, the limitations include sensitivity to lighting conditions and image quality, the requirement for large labeled datasets, and the inability to fully capture attention or interaction behavior.

2.3 EYE GAZE TRACKING APPROACHES

Eye gaze behavior is a strong indicator of ASD, as individuals with autism often show reduced attention to social stimuli such as faces and eye contact.

A study by Jaradat et al. (2024) used eye-tracking datasets to develop a hybrid machine learning model that achieved up to 98% accuracy in ASD classification. Similarly, Alsharif et al. (2024) proposed a deep learning-based eye-tracking system that achieved near-perfect accuracy in distinguishing autistic and non-autistic individuals. A systematic review conducted in 2025 reported that machine learning models using eye-tracking data achieved an average accuracy of around 85–86% with high discriminative ability (AUC ≈ 0.92).

Eye-tracking systems measure:

- Fixation duration
- Gaze direction
- Attention to social vs non-social stimuli

These features provide valuable insights into cognitive and social behavior.

In general, the advantages of these approaches include their strong behavioral relevance, objectivity, and high diagnostic accuracy. However, the limitations include the need for real-time processing, sensitivity to environmental conditions, and the requirement for proper calibration and setup

2.4 DEEP LEARNING AND COMPUTER VISION APPROACHES

Deep learning has significantly improved ASD detection by enabling automatic feature extraction from complex data. Instead of relying on manually designed features, these models learn directly from raw inputs such as facial images and behavioral videos, capturing subtle patterns that are difficult to observe visually. Recent advancements in deep learning architectures have made it possible to analyze both spatial and temporal characteristics of data, which is particularly useful for identifying ASD-related behavioral traits. The use of large datasets along with pre-trained models has further enhanced model performance by improving generalization and reducing training time. As a result, deep learning has become a key approach for building automated ASD screening systems that support early detection.

Recent studies have explored hybrid architectures combining CNN, LSTM, and Transformer models to capture both spatial and temporal features. CNNs are effective in extracting spatial features from images, such as facial structures and textures, while LSTMs help in modeling sequential patterns across time. Transformer models, with their attention mechanisms, improve the ability to focus on the most relevant features within the input data. By combining these models, hybrid systems can better analyze facial expressions, gaze behavior, and emotional responses over time. This integrated approach provides a more comprehensive understanding of behavioral patterns associated with ASD and improves classification accuracy compared to single-model approaches.

Computer vision-based systems have also been developed to analyze child interactions, posture, and activity patterns using video data. These systems can detect behavioral indicators such as reduced eye contact, repetitive movements, and atypical social engagement by processing continuous frames. Techniques like pose estimation and action recognition are used to extract meaningful features from dynamic environments. When combined with deep learning models, these systems can automatically interpret complex behaviors without manual intervention. Overall, deep learning approaches offer high accuracy and automatic feature learning, but they also come with limitations such as high computational requirements, dependence on large labeled datasets, and reduced interpretability due to their black-box nature.

2.5 MULTIMODAL ASD DETECTION SYSTEMS

Recent research emphasizes the importance of combining multiple modalities to improve detection accuracy.

A multimodal study conducted in 2025 analyzed facial expressions, gaze behavior, speech, and physiological signals together, showing improved classification performance compared to single-modality systems. Another approach combined Vision Transformers with eye-tracking and speech data, achieving high accuracy of around 96% using attention-based fusion techniques.

Multimodal systems typically use feature-level fusion, decision-level fusion, or weighted ensemble methods to integrate different data sources. The advantages of these systems include higher accuracy, comprehensive analysis, and better generalization. However, the limitations include complex system design, high computational requirements, and challenges in integrating heterogeneous data effectively.

2.6 COMPARATIVE ANALYSIS OF EXISTING METHODS

The following table presents a comparative analysis of different methodologies used for autism spectrum disorder detection, highlighting their input types, performance levels, advantages, and limitations.

Method	Input Type	Accuracy	Advantages	Limitations
ML Models	Questionnaire/Data	Moderate	Simple, fast	No behavioral analysis
Facial Analysis (CNN)	Images	Moderate	Non-invasive	Sensitive to image quality
Eye Tracking	Gaze Data	Very High (~85–98%)	Strong behavioral indicator	Needs real-time setup
Deep Learning	Video/Image	High	Automatic feature extraction	High cost
Multimodal Systems	Multiple Inputs	Very High (~90%+)	Best performance	Complex integration

Table 2.1 : Comparative Analysis of Existing Methods

2.7 RESEARCH GAPS IDENTIFIED

Based on the reviewed literature, the following research gaps have been identified:

- **Lack of integrated multimodal systems**
  Many studies focus on a single modality, reducing overall reliability.

- **Limited real-time web-based implementations**
  Most models are tested in controlled environments and not deployed for public use.

- **Inefficient fusion techniques**
  Existing systems do not effectively combine multiple data sources.

- **High computational complexity**
  Advanced models require powerful hardware, limiting accessibility.

- **Absence of automated report generation**
  Most systems do not provide interpretable outputs for users.

- **Limited accessibility in rural areas**
  Many solutions are not designed for low-resource environments.

2.8 SUMMARY OF LITERATURE SURVEY

The literature survey highlights that significant progress has been made in the field of Autism Spectrum Disorder (ASD) detection using computational techniques. Various approaches, including machine learning, deep learning, facial image analysis, eye gaze tracking, and multimodal systems, have been explored to improve early screening and diagnostic accuracy.

Machine learning-based methods are effective for structured datasets and provide efficient and interpretable solutions, but they are limited in handling complex behavioral patterns. Deep learning and computer vision approaches overcome this limitation by automatically extracting features from images and videos, achieving higher accuracy. Facial image-based detection methods have shown promising results due to their non-invasive nature, while eye gaze tracking techniques provide strong behavioral insights and high diagnostic relevance.

CHAPTER 3: PROPOSED SYSTEM

3.1 SYSTEM OVERVIEW

The proposed system is designed as a multimodal screening platform for Autism Spectrum Disorder (ASD), integrating multiple sources of information to improve the reliability of risk assessment. Instead of relying on a single method, the system combines behavioral, visual, and cognitive indicators to capture a broader representation of user characteristics.

The system interacts with the user through a structured interface that enables the collection of different forms of input, including questionnaire responses, facial images, and gaze-related data. These inputs represent complementary aspects of ASD-related behavior, allowing the system to analyze both observable traits and underlying cognitive patterns. Each type of data is processed using appropriate computational techniques, ensuring that the unique characteristics of each modality are effectively utilized.

The behavioral component focuses on identifying patterns in user responses using established scoring mechanisms, while the visual component leverages deep learning models to extract meaningful features from facial images. Together, these components contribute independent yet interconnected perspectives on ASD-related traits.

A key aspect of the system is the integration of these heterogeneous data sources through a fusion mechanism. This approach enables the system to combine multiple prediction outputs into a unified decision, improving overall accuracy and robustness. By considering correlations between different modalities, the system reduces the limitations associated with individual methods and enhances the reliability of the final assessment.

The output of the system is presented in the form of an interpretable risk classification, supported by a detailed report that summarizes the findings. The design emphasizes clarity and usability, ensuring that users can easily understand the results while maintaining a structured and consistent workflow.

Overall, the proposed system provides a scalable and efficient framework for ASD screening by combining multiple analytical approaches into a single unified platform. This integration not only improves prediction performance but also makes the system more adaptable to real-world applications.

3.2 SYSTEM ARCHITECTURE

The system is designed using a modular architecture in which each component is responsible for a specific function. This approach improves flexibility and allows easy modification or extension of the system in the future without affecting the overall structure. By dividing the system into independent modules, it becomes easier to manage, maintain, and upgrade individual components.

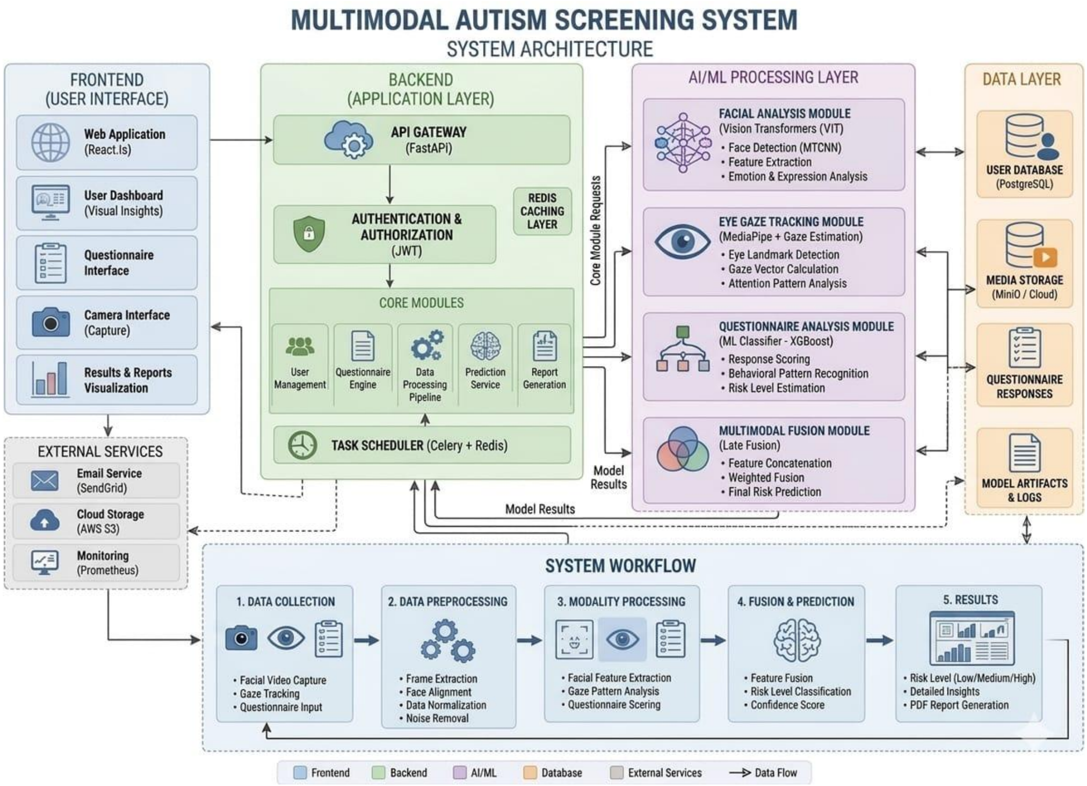

Figure 3.1: System architecture of multimodal system

The user interaction is handled through a dedicated interface that allows users to provide inputs and view results in a clear and understandable format. This layer ensures smooth communication between the user and the system while maintaining simplicity and usability. It is designed to be intuitive so that users can easily navigate through the system without requiring technical knowledge. The interface also validates user inputs to ensure that only correct and meaningful data is processed. This helps in reducing errors and improving the overall reliability of the system.

The core functionality of the system is carried out by the backend processing components, where all major computations take place. Different modules are responsible for handling specific types of data, including questionnaire evaluation, facial analysis, and gaze tracking. Each of these modules processes the input independently and extracts meaningful features related to ASD detection.

The outputs generated from these modules are then integrated using a fusion mechanism. This component combines the individual results into a single unified score, ensuring that all input sources contribute to the final decision. This integration improves the overall accuracy and reliability of the system.

In addition, a database component is included to securely store user data, intermediate outputs, and final results. This ensures data consistency, enables future analysis, and supports system scalability.

Overall, the modular structure ensures that each part of the system functions independently while still contributing effectively to the complete system workflow, resulting in an organized, efficient, and scalable architecture. This structure also allows the system to be easily extended with new features in the future without major redesign.

3.3 SYSTEM DESIGN

The system design focuses on how inputs are collected, processed, and transformed into meaningful outputs. It ensures that the system operates in a structured and efficient manner by organizing the workflow into well-defined components.

3.3.1 Input Design

The input design is responsible for collecting various types of data required for ASD screening. The system accepts basic user details such as age and gender, along with responses to behavioral questionnaires that capture social and cognitive patterns. In addition to textual inputs, the system also allows users to upload facial images and provide real-time video data for gaze tracking analysis. The interface is designed to be simple and user-friendly, ensuring that users can provide the required information without confusion or difficulty. Proper input validation mechanisms are also considered to maintain data consistency and reliability. This helps in preventing invalid or incomplete data from being processed by the system. It also ensures that all inputs are standardized before being passed to the backend modules for further analysis.

3.3.2 Processing Design

The processing design handles the transformation of raw input data into structured and analyzable formats. Each type of input is processed using appropriate techniques suited to its nature. Facial images undergo preprocessing steps such as resizing and normalization to ensure compatibility with deep learning models. Video data is analyzed frame by frame to detect and track eye movements, enabling the extraction of gaze-related features. Behavioral questionnaire responses are evaluated using predefined rule-based scoring methods. This stage ensures that all inputs are converted into meaningful representations that can be effectively used for further analysis and decision-making.

3.3.3 Output Design

The output design focuses on presenting the results of the analysis in a clear and understandable manner. The system generates a numerical risk score that indicates the likelihood of ASD, along with a categorized interpretation such as low, medium, or high risk. Additional supporting information is also provided to help users better understand the results and their implications. The output interface is designed to be intuitive and accessible, ensuring that even non-technical users can easily interpret the findings without requiring specialized knowledge.

3.4 MODULE DESCRIPTION

The system is divided into several modules, each responsible for a specific task.

3.4.1 User Interface and Consent Module

This module serves as the entry point of the system. It collects user consent and ensures that all data is handled securely.

It also gathers basic information required for analysis. The interface is designed to be simple and interactive, allowing users to easily navigate through different steps**.**

3.4.2 Questionnaire Assessment Module

This module evaluates behavioral patterns using standardized questionnaires. Each response is assigned a score, and the total score reflects the likelihood of autism-related traits. The questionnaire is designed to capture key behavioral indicators such as social interaction, communication ability, repetitive behaviors, and attention patterns. Based on the user's responses, the module applies a predefined scoring mechanism to quantify the results in a structured manner. The aggregated score is then compared against a threshold to determine the level of ASD risk. This approach ensures a systematic and consistent evaluation of behavioral characteristics.

This module is important because behavioral assessment is one of the most reliable methods for identifying ASD. Questionnaires are widely used in clinical settings as a preliminary screening tool due to their simplicity and effectiveness in capturing observable traits.

3.4.3 Facial Analysis Module

The facial analysis module processes images using deep learning techniques. The image is first preprocessed and then passed through a trained model.

The model analyzes features such as facial expressions and patterns that may be associated with autism. It produces a probability score indicating the likelihood of ASD.

This method is non-invasive and provides valuable visual insights.

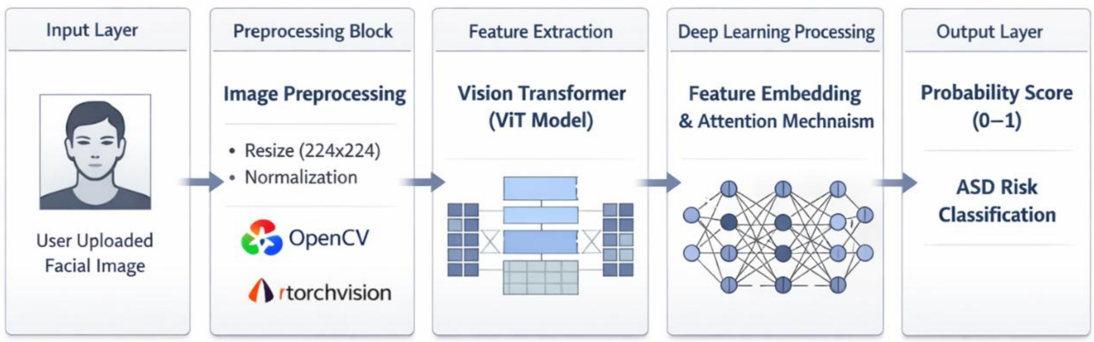

Figure 3.2: Facial Image-Based ASD Detection Pipeline Using Vision Transformer (ViT) for Automated Feature Extraction and Classification

3.4.4 Gaze Tracking Module

The gaze tracking module analyzes eye movement patterns using real-time video input by detecting the user's eyes and continuously monitoring their gaze direction. It identifies regions of interest within the frame and tracks how the gaze shifts over time, allowing the system to understand where the user is focusing during interaction. By measuring the duration and frequency of gaze fixation on specific areas, the module provides insights into attention behavior and visual engagement patterns. This information is particularly useful in identifying atypical gaze behavior, which is often associated with ASD. A key metric used in this module is the Social Preference Index (SPI), which represents the proportion of time spent looking at social stimuli compared to non-social elements, helping quantify social attention levels in a structured manner.

$$SPI = Ts/Tt$$

The Social Preference Index (SPI) is calculated as the ratio of time spent on social stimuli to the total viewing time. In this equation, $\boldsymbol{T_s}$represents the duration for which the user focuses on social elements such as faces and eye regions, while $\boldsymbol{T_t}$denotes the total time spent observing all stimuli, including both social and non-social elements. This metric provides an indication of the user's attention towards social cues, where higher values correspond to increased social engagement and lower values may indicate reduced attention to social stimuli.

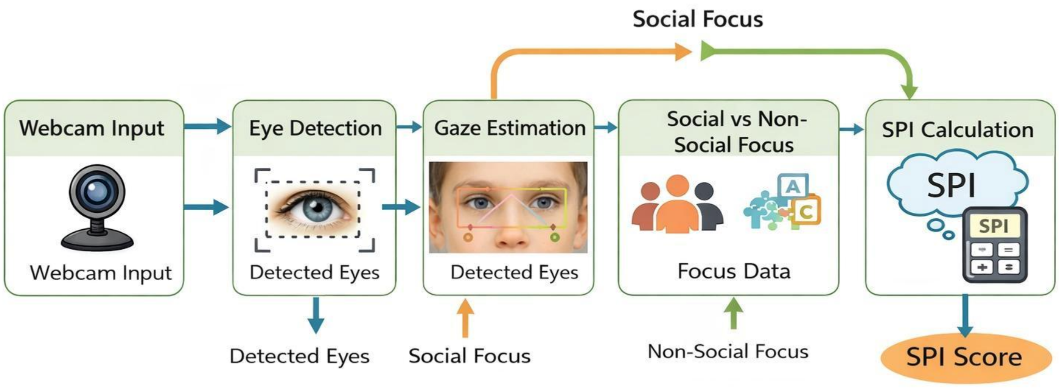

Figure 3.3: Gaze Tracking Module Pipeline for ASD Detection

The gaze tracking pipeline begins with real-time video input, where frames are captured and processed to detect the user’s face and eye regions. Once the eyes are localized, the system tracks eye movements across consecutive frames to determine the direction of gaze. Key features such as fixation points, gaze duration, and movement patterns are extracted to analyze visual attention behavior. These extracted features are then used to compute metrics like the Social Preference Index (SPI), which helps quantify the user’s focus on social versus non-social stimuli, contributing to ASD assessment.

3.4.5 Multimodal Fusion Module

This module is responsible for integrating the outputs obtained from the different components of the system. Since each module captures distinct aspects of user behavior, combining these outputs helps in improving the overall accuracy and reliability of the final prediction. To achieve this, a weighted fusion approach is adopted, where each module contributes to the final score based on its relative importance. This strategy ensures that the strengths of individual modules are effectively utilized while minimizing the impact of any single unreliable or noisy input. The fusion process is designed to maintain consistency across varying input conditions and ensures that no single modality dominates the final decision unfairly. It also allows the system to adapt to variations in input quality from different modules.

$$R=0.6Q+0.3G+0.1F$$

In this equation, R represents the final risk score, while Q, G, and F correspond to the outputs of the questionnaire module, gaze tracking module, and facial analysis module, respectively. The assigned weights indicate the contribution of each component to the overall decision, with higher weights given to more reliable or informative inputs. The questionnaire module is given the highest weight due to its strong clinical relevance in behavioral assessment, while gaze tracking contributes moderately by providing measurable attention patterns. Facial analysis, though useful, is assigned a comparatively lower weight due to its sensitivity to external factors such as lighting and image quality. This weighted combination allows the system to balance multiple sources of information in a structured manner.

This approach ensures that the final result reflects a balanced and optimized combination of all available data sources, thereby enhancing the effectiveness of the ASD risk assessment. It also improves robustness by reducing dependency on any single module, making the system more stable across different input conditions. Additionally, the fusion mechanism supports scalability, as new modules or data sources can be incorporated by adjusting the weights accordingly. Overall, this method provides a systematic way to integrate multimodal outputs and generate a more accurate and reliable final prediction. It further enables extensibility, allowing future improvements in individual modules to directly enhance the overall system performance without requiring major architectural changes.

3.4.6 Report Generation Module

The report generation module is responsible for producing the final output of the system in a structured and interpretable format. It consolidates the processed results obtained from various modules, including questionnaire evaluation, facial analysis, and gaze tracking, and presents them as a unified assessment. The module transforms the computed risk score into meaningful information by categorizing the autism risk level and associating it with clear descriptions. This ensures that the output is not merely numerical but also informative and actionable. The report is generated after all modules complete their processing, ensuring that the final output reflects the combined insights from all data sources. It acts as the concluding stage of the system pipeline, where all intermediate computations are summarized into a coherent and user-friendly format while maintaining consistency in presentation.

In addition to presenting the overall risk level, the module includes supporting insights derived from individual components of the system. These details help users understand how different factors contributed to the final result, thereby increasing transparency and trust in the system. The report may include a breakdown of scores from each module, such as questionnaire results, gaze tracking metrics like the Social Preference Index, and facial analysis outputs, along with their respective contributions to the final score. It may also incorporate visual elements such as charts or graphs to enhance readability and make the interpretation more intuitive. This structured representation allows users to easily compare module-wise outputs and gain a clearer understanding of the underlying analysis.

The report is designed with a focus on clarity and accessibility, ensuring that even users without technical knowledge can interpret the findings effectively. Additionally, it can include basic user details and timestamps for record-keeping, along with optional recommendations suggesting further evaluation by a specialist if higher risk levels are detected. The module also ensures that the generated report can be stored, viewed later, or exported in formats such as PDF for documentation purposes. By providing a comprehensive yet concise summary, the module assists users in making informed decisions and supports effective communication of the system's results while maintaining usability and consistency.

3.5 IMPLEMENTATION DETAILS

The system is implemented using a combination of modern web technologies and machine learning frameworks to ensure both performance and scalability. The frontend is developed using React, which provides a responsive and interactive user interface for seamless user interaction. This enables efficient data collection and visualization while maintaining a smooth user experience across different devices. The backend is implemented using Python and FastAPI, which supports high-performance API development and enables efficient communication between different system components.

The machine learning and computer vision functionalities are supported by libraries such as OpenCV, MediaPipe, and PyTorch. OpenCV is utilized for image preprocessing and video frame handling, while MediaPipe facilitates real-time tracking of facial landmarks and eye movements. PyTorch is employed for implementing deep learning models used in facial analysis and pattern recognition. The integration of these technologies ensures efficient data processing, real-time responsiveness, and the ability to handle complex computations. Overall, the implementation framework is designed to balance performance, flexibility, and ease of deployment in real-world environments.

3.6 WORKING OF THE SYSTEM

The working of the system is based on a coordinated interaction between frontend interfaces, backend processing modules, and machine learning components, forming a complete end-to-end screening pipeline. The system collects multiple forms of input from the user and processes them through specialized modules, each designed to capture a specific aspect of behavior relevant to Autism Spectrum Disorder (ASD). These modules operate independently but are integrated through a centralized workflow to produce a unified assessment.

The questionnaire module functions as the primary screening component, where users respond to standardized tools such as AQ-10 or Social Communication Questionnaire (SCQ). A scoring algorithm is applied, including reverse scoring for certain questions, to compute a total score. Based on predefined thresholds, the system classifies the result into low, medium, or high risk categories . This module forms the most significant component of the system due to its clinical relevance.

The facial analysis module processes user-uploaded images using computer vision and deep learning techniques. The image is handled entirely in memory for privacy, and OpenCV is used to perform face detection and quality checks such as brightness and resolution. Once validated, the image is transformed using torchvision operations such as resizing to 224×224 pixels and normalization before being passed to a pre-trained Vision Transformer (ViT) model . The model outputs a probability score between 0 and 1, which is mapped to risk categories using threshold values. The processed image is immediately deleted from memory after analysis to ensure data privacy .

The gaze tracking module provides behavioral insights by analyzing eye movement patterns in real time. The system uses a webcam-based approach where the user undergoes a 9-point calibration process to train a personalized gaze estimation model . MediaPipe Face Mesh is used to detect facial landmarks, including 468 key points, enabling precise extraction of eye features. During the analysis phase, the system displays a split-screen video with social stimuli on one side and geometric patterns on the other. The system continuously captures frames and predicts gaze coordinates using the EyeTrax-based model, tracking whether the user is focusing on social or non-social content.

Based on the collected gaze data, the system computes the Social Preference Index (SPI), defined as the normalized difference between time spent on social and geometric stimuli. The SPI value is then used to determine the corresponding risk category using predefined thresholds, where higher values indicate stronger social preference and lower values suggest potential ASD-related traits . This module provides an objective behavioral signal that complements the other components.

The outputs from all modules are then combined using a risk fusion mechanism implemented in the backend. This module uses a weighted approach, assigning 60% importance to questionnaire results, 30% to gaze tracking, and 10% to facial analysis, reflecting their relative clinical significance . The risk categories are converted into numerical values and aggregated using a weighted average, followed by conversion back into a final risk category. The system also calculates a confidence score and generates an interpretation along with recommendations.

Finally, the results are presented to the user through a React-based interface, where visual components such as charts and risk indicators are used for better understanding. A detailed report is generated, including individual module outputs and the final assessment. This integrated workflow ensures accurate, real-time, and user-friendly ASD screening.

3.7 ADVANTAGES OF THE PROPOSED SYSTEM

The proposed system offers several advantages by effectively addressing the limitations associated with traditional Autism Spectrum Disorder (ASD) screening methods. One of the most significant benefits is its ability to provide faster and automated screening. Conventional diagnostic procedures often involve multiple stages of evaluation, including clinical observation, interviews, and standardized testing, which can be time-consuming. In contrast, the proposed system leverages artificial intelligence and machine learning techniques to perform preliminary assessments in a much shorter duration. This reduction in screening time enables quicker identification of potential ASD traits, allowing timely intervention and support.

Another key advantage is the reduced dependency on expert intervention during the initial stages of diagnosis. In many regions, especially rural and underdeveloped areas, access to trained specialists such as child psychologists and neurologists is limited. The proposed system bridges this gap by offering an intelligent screening mechanism that can be used without continuous expert supervision. While it does not replace professional diagnosis, it acts as an effective decision-support tool that can guide parents, caregivers, and educators toward seeking appropriate medical attention when necessary.

The integration of multiple data modalities significantly enhances the accuracy and reliability of the system. Unlike traditional approaches that may rely solely on behavioral observation or questionnaire-based assessments, this system combines various inputs such as facial image analysis, gaze tracking, and questionnaire responses. By analyzing behavioral, visual, and cognitive patterns simultaneously, the system is capable of identifying subtle indicators of ASD that might otherwise be overlooked. This multimodal approach reduces the chances of misclassification and improves the overall robustness of the screening process.

Accessibility and scalability are also major strengths of the proposed system. Being implemented as a web-based platform, it can be accessed from any location with an internet connection. This eliminates the need for specialized clinical infrastructure and allows users to perform preliminary screening from the comfort of their homes. Such accessibility is particularly beneficial in remote or underserved areas where healthcare facilities may be scarce. Furthermore, the system is designed to handle multiple users simultaneously, making it scalable for large-scale deployment in schools, clinics, and community health programs.

Cost-effectiveness is another important advantage. Traditional ASD diagnostic procedures can be expensive due to repeated consultations, specialized tests, and long-term evaluations. The proposed system minimizes these costs by providing an affordable initial screening solution. By identifying potential cases early, it reduces unnecessary clinical visits and allows healthcare resources to be allocated more efficiently. This makes the system economically beneficial for both families and healthcare providers.

The system also benefits from a modular and flexible architecture, which allows for easy updates and future enhancements. As advancements continue in fields such as deep learning, computer vision, and behavioral analysis, new modules or improved algorithms can be integrated without redesigning the entire system. This adaptability ensures that the system remains relevant and continues to improve over time in response to new research findings and technological innovations.

In addition, the system supports real-time or near real-time analysis for certain components such as gaze tracking, enhancing user interaction and responsiveness. Efficient implementation and optimized algorithms ensure that the platform remains user-friendly and performs well even in standard computing environments. This contributes to a smoother user experience and encourages wider adoption.

Finally, the system emphasizes data privacy and ethical considerations. Sensitive user data, including behavioral responses and facial images, are handled using secure storage and processing techniques. Proper user consent mechanisms are incorporated to ensure ethical data usage, which helps build trust among users and promotes responsible deployment of the technology.

Overall, the proposed system presents a comprehensive, efficient, and accessible solution for early ASD screening. By combining automation, multimodal analysis, scalability, and cost-effectiveness, it significantly improves upon traditional methods and has the potential to make early autism detection more practical and widely available.

3.8 APPLICATIONS

The proposed system has wide-ranging applications across multiple domains, particularly in healthcare and education. In healthcare settings, it can be used as an initial screening tool to assist professionals in identifying individuals who may require further clinical evaluation. By providing early indications of ASD risk, the system supports timely intervention, which is crucial for improving developmental outcomes. It can also be integrated into telemedicine platforms, enabling remote assessment and expanding access to diagnostic support.

The system provides an accessible and easy-to-use preliminary screening tool that enables families to identify potential developmental concerns at an early stage, even without immediate access to medical experts. This early awareness empowers parents to seek timely intervention and appropriate support when needed.

In educational environments, the system can be utilized for early identification of children who may be at risk, allowing educators and caregivers to take appropriate measures. Additionally, the system supports home-based preliminary assessment, enabling parents to monitor behavioral patterns in a convenient and non-invasive manner. Its adaptability to different use cases makes it a versatile tool for improving awareness, accessibility, and early detection of ASD across various contexts.

3.9 SUMMARY

A detailed description of the design and implementation of the proposed autism screening system has been presented, emphasizing the integration of multiple data sources such as behavioral, visual, and gaze-based inputs to ensure a comprehensive and reliable assessment. The modular architecture enables organized system functionality, where each component operates independently while contributing to the overall workflow, thereby improving maintainability, testing, and future enhancements. The use of advanced technologies, including machine learning and deep learning, enhances the system's ability to analyze complex patterns and detect subtle indicators of ASD, while performance optimization ensures smooth operation in a web-based environment.

By combining multiple analytical approaches within a unified framework, the system overcomes the limitations of traditional screening methods and provides a more holistic and accurate solution. Its scalable and user-friendly design allows deployment across healthcare, educational, and home settings, improving accessibility, especially in remote and underserved areas. Overall, the proposed approach demonstrates strong potential for delivering efficient, reliable, and accessible ASD screening, supporting early detection and timely intervention to improve developmental outcomes.

CHAPTER 4: RESULTS AND DISCUSSION

4.1 INTRODUCTION

This chapter presents the experimental results and performance evaluation of the proposed multimodal autism screening system. The system integrates questionnaire analysis, facial feature extraction, and gaze tracking to determine the risk of Autism Spectrum Disorder (ASD).

The main objective of this chapter is to evaluate the effectiveness of the system under different conditions and analyze how each module contributes to the final decision. The results are discussed using standard evaluation metrics such as accuracy, precision, recall, and F1-score.

In addition to numerical evaluation, this chapter also provides a detailed discussion of system behavior, module performance, and practical observations obtained during testing.

4.2 EXPERIMENTAL SETUP

The evaluation of the system was conducted in a controlled environment as well as through real-time user interaction. This ensures that the system performance reflects both theoretical accuracy and practical usability.

4.2.1 Software Environment

- Programming Language: Python
- Backend Framework: FastAPI
- Frontend Framework: React
- Machine Learning Libraries: PyTorch, Transformers
- Computer Vision Tools: OpenCV, MediaPipe

These tools were selected for their efficiency, scalability, and compatibility with real-time applications. Python serves as the primary programming language due to its simplicity, readability, and extensive support for scientific computing and machine learning tasks. It provides a rich ecosystem of libraries that facilitate rapid development and integration of complex algorithms required for ASD detection.

FastAPI is used as the backend framework because of its high performance, asynchronous capabilities, and ease of building RESTful APIs. It enables efficient communication between the frontend and backend components while ensuring low latency in handling requests. React is chosen for the frontend due to its component-based architecture, which allows the development of dynamic and responsive user interfaces. It enhances user experience by providing real-time interaction and smooth rendering of data-driven components.

For machine learning tasks, PyTorch and Transformers are utilized to build and deploy deep learning models. PyTorch offers flexibility and strong support for neural network development, while the Transformers library provides pre-trained models that can be fine-tuned for specific tasks such as image and sequence analysis. These libraries help in implementing advanced architectures efficiently with reduced development effort.

In terms of computer vision, OpenCV and MediaPipe are employed for processing images and video data. OpenCV is used for tasks such as image preprocessing, face detection, and frame manipulation, while MediaPipe provides ready-to-use pipelines for face mesh detection and landmark extraction. Together, these tools enable accurate tracking of facial features and eye movements in real time, which are essential for gaze tracking and facial analysis modules. Overall, the selected software environment ensures a robust, scalable, and efficient system capable of handling multimodal data for ASD detection.

4.2.2 Hardware Environment

The system was tested on a standard computing setup with the following configuration:

- Processor: Intel i5 / equivalent
- RAM: 8 GB
- Camera: Built-in or external webcam
- Operating System: Windows/Linux

The system does not require high-end hardware, making it suitable for general users. This ensures that the application can run efficiently on commonly available devices without significant performance issues. Additionally, the lightweight nature of the implementation allows smooth execution of modules such as real-time gaze tracking and facial analysis.

4.2.3 Dataset Description

The evaluation involved different types of data collected and utilized to train, validate, and test the proposed ASD detection system. Since the system is multimodal in nature, each module relies on a specific type of dataset corresponding to its functionality. The integration of these datasets enables the system to analyze ASD-related characteristics from multiple perspectives, including behavioral responses, facial features, and gaze patterns. By combining heterogeneous data sources, the system aims to achieve a more comprehensive and reliable assessment compared to single-modality approaches.

Facial Image Dataset: The facial image dataset contains labeled images of autistic and non-autistic children, which are used for training and validating the facial analysis module. These images are typically collected from publicly available datasets and research repositories, ensuring diversity in terms of age, gender, lighting conditions, and facial expressions. The dataset undergoes preprocessing steps such as resizing, normalization, and augmentation to improve model generalization and robustness. Data augmentation techniques like rotation, flipping, and scaling are applied to increase dataset variability and reduce overfitting. The labels associated with each image indicate whether the subject belongs to the ASD or non-ASD category, enabling supervised learning for classification tasks. This dataset plays a crucial role in enabling the deep learning model to learn discriminative facial features associated with ASD-related traits.

Questionnaire Dataset: The questionnaire dataset is based on standardized screening tools such as AQ-10 (Autism Spectrum Quotient) and SCQ (Social Communication Questionnaire), which are widely used in clinical and research settings. This dataset consists of structured responses collected from users through a series of behavioral questions that assess social interaction, communication skills, attention patterns, and repetitive behaviors. Each response is assigned a numerical value, and the aggregated responses form a feature vector representing the behavioral profile of the individual. The dataset is used by the questionnaire assessment module to compute a total score that reflects the likelihood of ASD traits. Since these questionnaires are validated instruments, they provide reliable and interpretable behavioral indicators that complement other data sources in the system.

Gaze Data: Gaze data is collected in real time using webcam-based eye tracking during system interaction. This dataset captures eye movement patterns, including gaze direction, fixation duration, and transitions between different regions of interest on the screen. The system processes video frames using computer vision techniques to detect facial landmarks and track eye positions continuously. From this data, features such as gaze duration, frequency of attention shifts, and Social Preference Index (SPI) are derived. Unlike static datasets, gaze data is dynamic and temporal in nature, requiring sequential analysis to extract meaningful behavioral insights. This dataset is particularly valuable as it reflects natural user behavior in an interactive environment, making it suitable for behavioral analysis related to ASD.

The combination of these datasets allows comprehensive testing of all system modules. Each dataset contributes unique and complementary information, enabling the system to analyze ASD from multiple dimensions. While the facial image dataset captures visual and structural features, the questionnaire dataset provides behavioral insights, and the gaze data reflects real-time attention and interaction patterns. Together, these datasets support the development of a robust multimodal system that improves prediction accuracy and generalization. Additionally, the integration of diverse data sources ensures that the system is not overly dependent on a single modality, thereby enhancing reliability across different scenarios and user inputs.

4.3 EVALUATION METRICS

To measure system performance, standard classification metrics were used. These metrics help in understanding how accurately the system predicts ASD risk by comparing predicted outputs with actual ground truth values. The evaluation is based on four key terms commonly used in classification problems:

- True Positive (TP): Correctly predicted positive cases (ASD cases correctly identified as ASD).
- True Negative (TN): Correctly predicted negative cases (non-ASD cases correctly identified as non-ASD).
- False Positive (FP): Incorrectly predicted positive cases (non-ASD cases wrongly classified as ASD).

- False Negative (FN): Missed positive cases (ASD cases wrongly classified as non-ASD).

Using these values, different performance measures are computed to evaluate the effectiveness of the system.

4.3.1 Accuracy

Accuracy represents the overall correctness of the system by measuring the proportion of correctly classified instances among all predictions. It is calculated as:

Accuracy = (TP + TN) / (TP + TN + FP + FN)

A higher accuracy indicates that the model is correctly predicting both ASD and non-ASD cases in a balanced manner. However, accuracy alone may not be sufficient in cases where the dataset is imbalanced, as it does not distinguish between types of errors.

4.3.2 Precision

Precision Precision measures how many of the predicted positive cases are actually correct. It focuses on the reliability of positive predictions made by the system. It is calculated as:

Precision = TP / (TP + FP)

High precision reduces the chances of false alarms, meaning fewer non-ASD cases are incorrectly classified as ASD. This metric is important when the cost of false positives needs to be minimized.

4.3.3 Recall

Recall indicates how effectively the system identifies actual ASD cases. It measures the proportion of correctly identified positive cases out of all actual positive cases. It is calculated as:

$$Recall = \frac{TP}{TP + FN}$$

A high recall value means the system successfully detects most of the true ASD cases, which is particularly important in medical screening applications where missing a positive case can have serious consequences.A high recall value means the system successfully detects most of the true ASD cases, which is particularly important in medical screening applications where missing a positive case can have serious consequences.

4.3.4 F1-Score

F1-score provides a balance between precision and recall by combining both metrics into a single value. It is especially useful when there is a trade-off between false positives and false negatives. It is calculated as:

$$F1 = \frac{2PR}{P + R}$$

This metric is useful when both types of errors need to be minimized, providing a more comprehensive evaluation of the model's performance compared to using precision or recall alone.

4.4 OVERALL SYSTEM PERFORMANCE

The performance of the system was evaluated after integrating the available modules.

Observed Performance Values:

Metric	Value
Accuracy	~80%
Precision	0.78 – 0.82
Recall	0.72 – 0.78
F1-Score	~0.75

Table 4.1 : Performance Evaluation Metrics of the Proposed ASD Detection System

These results indicate that the system performs reasonably well in its current stage of development.

The accuracy is primarily influenced by the questionnaire module, while the facial and gaze modules are still undergoing optimization. Future improvements in these modules and the incorporation of larger and more diverse datasets are expected to further enhance the overall performance and robustness of the system. Additionally, fine-tuning the model parameters and fusion weights may lead to better balance across all evaluation metrics.

4.5 MODULE-WISE ANALYSIS

4.5.1 Questionnaire Module Performance

The questionnaire module demonstrated the most stable and consistent performance among all modules.

- Provides reliable behavioral assessment
- Based on clinically validated methods
- Low computational cost

Since this module is rule-based, its output remains consistent across different inputs, making it a strong contributor to the final decision**.**

4.5.2 Facial Analysis Module Performance

The facial analysis module showed moderate performance during testing.

- Accuracy depends on image quality
- Performance improves with better dataset training
- Sensitive to lighting and facial orientation

The module achieved an approximate accuracy of 80%, with prediction probabilities typically ranging between 0.3 and 0.7 depending on image conditions.Although the current performance is moderate, it is expected to improve significantly after further training and optimization of the model.

4.5.3 Gaze Tracking Module Performance

The gaze tracking module provides important behavioral insights.

- Works effectively in controlled environments
- Sensitive to camera position and lighting
- Provides dynamic behavioral data

This module enhances the system's ability to analyze attention patterns, which are important indicators of ASD.

4.6 MULTIMODAL FUSION ANALYSIS

The fusion module combines outputs from all individual modules using a weighted approach.

Module	Contribution
Questionnaire	High
Gaze Tracking	Medium
Facial Analysis	Low

Table 4.2: Contribution of Each Module

Currently, the system relies more on questionnaire data due to higher reliability. However, as other modules improve, the contribution will become more balanced.

The fusion approach ensures that weaknesses in one module are compensated by strengths in others.

4.7 DISCUSSION OF RESULTS

The results obtained demonstrate that the proposed system is capable of performing early autism screening with reasonable accuracy.

One of the key observations is that combining multiple modalities improves the robustness of the system. While individual modules have limitations, their integration provides a more comprehensive understanding of autism-related behavior.

The current accuracy of approximately 80% indicates that the system is functional and can be used for preliminary screening. However, further improvements are required to achieve clinical-level accuracy.

The system also demonstrates good scalability and flexibility due to its modular architecture, allowing individual components to be improved or replaced without affecting the overall workflow. This makes it easier to incorporate advanced models, larger datasets, or additional modalities in the future. Furthermore, the lightweight design of the questionnaire module combined with the real-time capabilities of the gaze and facial analysis modules ensures a balanced trade-off between accuracy and computational efficiency. With further optimization, data refinement, and real-world testing, the system has strong potential to evolve into a reliable and widely usable screening tool.

The system also performs efficiently in real-time conditions, making it suitable for practical applications. However, environmental factors such as lighting conditions, camera quality, and user interaction can influence performance.

Overall, the results validate the effectiveness of the multimodal approach and highlight the potential for further enhancement.

4.8 VISUALIZATION AND OUTPUT ANALYSIS

The system generates a visual report that includes:

- Risk level classification
- Individual module scores
- Graphical representation (charts)

These visual elements improve user understanding and make the results easier to interpret.

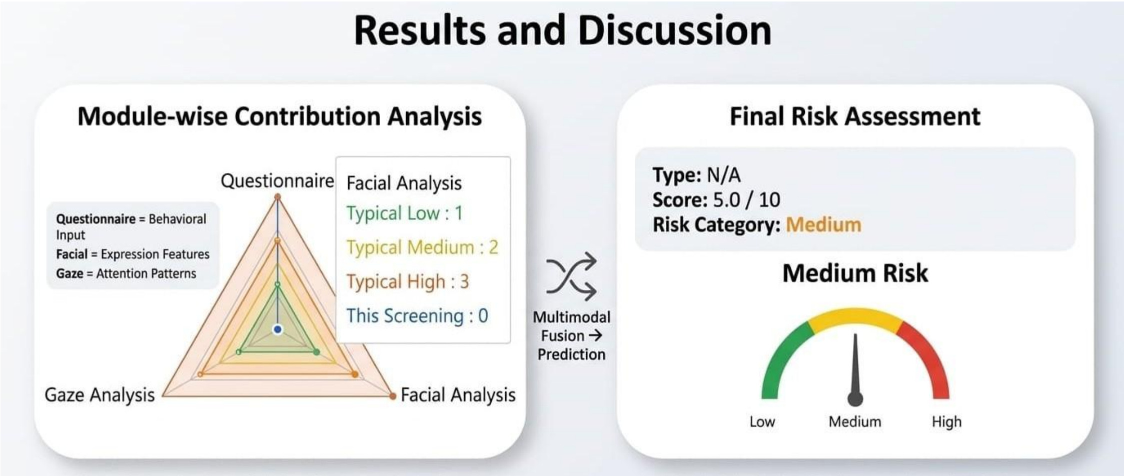

Fig 4.1: Combined Multimodal Risk Analysis and Final Autism Risk Prediction

The report is designed to be user-friendly, ensuring that even non-expert users can understand the outcome.

4.9 LIMITATIONS OF THE SYSTEM

Despite promising results, the system has certain limitations:

- Facial model requires further training
- Gaze tracking affected by environmental conditions
- Limited dataset size
- Dependence on input quality

Addressing these limitations will improve system performance in future versions.

4.10 SUMMARY

This chapter presented a detailed evaluation of the proposed system. The results indicate that the system achieves an accuracy of approximately 80% in its current stage.

The multimodal approach enhances reliability and provides a comprehensive assessment of autism risk. Although there are certain limitations, the system shows strong potential for further improvement and real-world application.

CHAPTER 5: CONCLUSIONS

5.1 CONCLUSION

In this project, a multimodal autism screening system was designed and developed to assist in the early detection of Autism Spectrum Disorder (ASD) in children. The system integrates multiple data sources, including behavioral questionnaire responses, facial analysis, and eye gaze tracking, to provide a comprehensive and reliable risk assessment. The primary objective of the system was to overcome the limitations of traditional diagnostic methods, which are often time-consuming, subjective, and dependent on expert availability. By utilizing artificial intelligence and machine learning techniques, the proposed system automates the screening process and provides faster and more objective results.

The system was implemented as a web-based platform, ensuring accessibility and ease of use for parents, educators, and healthcare professionals. The modular design of the system allows each component to function independently while contributing to the overall decision-making process.

From the experimental results, it was observed that the system achieved an accuracy of approximately 80% in its current stage. The questionnaire module provided the most stable performance, while the facial analysis and gaze tracking modules contributed additional behavioral insights. The use of a weighted fusion approach enabled effective integration of these modalities, improving the overall reliability of the system.

Although the system is still under development in certain aspects, the results demonstrate that the proposed approach is feasible and effective for preliminary autism screening. The system provides a strong foundation for further research and development in AI-based healthcare solutions.

5.2 KEY ACHIEVEMENTS

The major achievements of the project are summarized below:

- Successfully designed a multimodal screening system integrating multiple data sources
- Developed a web-based platform for easy accessibility

- Implemented machine learning models for facial analysis
- Designed a gaze tracking mechanism for behavioral analysis
- Integrated standardized questionnaires for reliable assessment
- Achieved real-time processing capability
- Generated automated reports for user-friendly interpretation

These achievements highlight the effectiveness of combining different technologies to solve real-world healthcare problems.

5.3 LIMITATIONS

Despite its advantages, the system has certain limitations that need to be addressed:

- The facial analysis module requires further training for improved accuracy
- Gaze tracking performance is affected by environmental conditions such as lighting and camera quality
- The dataset used is limited in size, which may impact model generalization
- The system is currently designed for screening and not for clinical diagnosis
- Performance may vary depending on user input quality

Addressing these limitations will enhance the system's reliability and usability.

5.4 FUTURE SCOPE

The proposed system has significant potential for further improvement and expansion. Several enhancements can be implemented to increase its accuracy, usability, and real-world applicability.

5.4.1. Integration of Additional Modalities

Future versions of the system can include additional inputs such as:

- Speech and audio analysis
- Body movement and gesture recognition
- Behavioral video analysis

These additions will provide a more comprehensive assessment of ASD.

5.4.2. Improved Machine Learning Models

Advanced models can be incorporated to improve accuracy:

- More powerful deep learning architectures
- Transformer-based multimodal models
- Improved training using larger datasets

This will enhance the system's ability to detect subtle patterns.

5.4.3. Mobile Application Development

The system can be extended into a mobile application to improve accessibility. This will allow users to perform screening using smartphones, making it more convenient and widely usable.

5.4.4. Multilingual Support

Adding support for multiple languages will make the system accessible to users from different regions and backgrounds. This is especially important for deployment in diverse populations.

5.4.5. Clinical Validation

Future work can involve collaboration with healthcare professionals to validate the system in clinical settings. This will help in improving accuracy and reliability for real-world applications.

5.4.6. Enhanced Data Collection

Collecting larger and more diverse datasets will improve model training and generalization. This will make the system more robust and adaptable to different conditions.

5.4.7. Real-Time Optimization

Improving system performance for real-time processing will enhance user experience. Optimization techniques can be applied to reduce latency and improve efficiency.

5.5 FINAL REMARK

The proposed multimodal autism screening system demonstrates the potential of artificial intelligence in improving early detection of developmental disorders. By combining multiple data sources and leveraging modern technologies, the system provides a promising solution for accessible and efficient ASD screening.

With further improvements and real-world validation, the system can become a valuable tool in supporting early diagnosis and intervention, ultimately contributing to better outcomes for children with autism. In addition, the system can assist healthcare professionals by providing preliminary insights, reducing manual effort, and enabling faster decision-making. Its ability to operate in a user-friendly and cost-effective manner also increases its suitability for deployment in remote or resource-limited settings, thereby expanding the reach of early screening services.

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